

Exercise 02: Bubble Sort

The core step of the bubble sort algorithm makes a single pass through a subarray from front to back, comparing each pair of consecutive elements and swapping them if the first is greater than the second, so that after the maximum value in the subarray has moved to its last position. Write a C++ program named 'main02.cpp' that satisfies the following requirements:

1. Include only the libraries 'iostream', 'string', 'cstdlib', 'ctime', and 'vector'.
2. Define the `print()` function,

```
void print(vector<int>& q)
```

that displays the elements of `q` as a comma-separated list enclosed in square brackets.

3. Define the `moveMax()` function,

```
void moveMax(vector<int>& q, int i, int j)
```

that performs the core bubble-sort step on the subarray `q` between `i` and `j` (inclusive), only if `i` and `j` are valid indices of `q`.

4. Testing in the `main()` function:
 - a. Initialize a vector object with random values.
 - b. Invoke `print()` to display the vector object.
 - c. Invoke `moveMax()` on the vector object with either user-provided or randomly generated indices.
 - d. Invoke `print()` again to display the vector object.
 - e. Repeat steps c and d with different indices.